REPORT

1.

Modeling overall hotel ratings [1-5] with review texts as well as review's title to predict the given overall ratings.

Texts are mined by indetifying which words appears with a frequency of 5 or more, and using a machine learning to construct a predictive model for ratings associated with spesific words.

RandomForest is used for robustness of the model and it is considered a good choice for text mining. Also, model uses cross-validtion of 10 fold to create a robust prediction.

Additionally a cost matrix analysis is also incurred in to the analysis - costs from false positives and false negatives for the business.

> -Assumption of extreme false positive (1->5) to be at least 4 times the cost as extreme false negative (5->1) from the lack of action taken, such as improvements and investments in the business. Cost of false negative arise e.g from the possibility of not being able to capture the profit from e.g forecasting future visits (assumption: visits correlates with hotel rating).

```
Cost Matrix
 0    2    4    6    8
 0.5  0    3.25 4.63 6
 1    1.75 0    3.25 4
 1.5  1.63 1.75 0    2
 2    1.5  1    0.5  0
```

Analysis

ZeroR, which works as a benchmark, as it assigns the instance to a rating that is most likely to appear in the dataset a.k.a mode (4.0). ZeroR produced 34.9583 % of correctly classified instances for both title and the text datasets.

```
Correctly Classified Instances        1090               34.9583 %
Incorrectly Classified Instances      2028               65.0417 %
Kappa statistic                          0
Total Cost                            4987.75
Average Cost                             1.5997
```

Title:

```
Correctly Classified Instances        1474               47.2739 %
Incorrectly Classified Instances      1644               52.7261 %
Kappa statistic                          0.2654
Total Cost                            4176.875
Average Cost                             1.3396
=== Confusion Matrix ===

   a   b   c   d   e   <-- classified as
 116  30  28  38  61 |   a = 1
  41  22  59  80  62 |   b = 2
  21  29 169 184 100 |   c = 3
  23  23 116 570 358 |   d = 4
  23  12  64 292 597 |   e = 5
```

Correctly classified instances is significally higher than benchmark, indicating predictive power of the model.

Attribute selection - >0:

```
Correctly Classified Instances        1447               46.408  %
Incorrectly Classified Instances      1671               53.592  %
Kappa statistic                          0.2238
Total Cost                            4122.75
Average Cost                             1.3222
=== Confusion Matrix ===

   a   b   c   d   e   <-- classified as
  95  19  11 126  22 |   a = 1
  26  17  48 151  22 |   b = 2
   3  25 110 329  36 |   c = 3
   5  13  57 809 206 |   d = 4
   7   7  11 547 416 |   e = 5
```

Parsing the data to include only attributes with information gain with respect to the class actually decreases predictive accuracy – but a small decrease in total cost. However, confusion matrix indicates, that extreme false positives and negatives are reduced as the noise is reduced.

CSC (all attributes):

```
Correctly Classified Instances        1071               34.3489 %
Incorrectly Classified Instances      2047               65.6511 %
Kappa statistic                          0.162
Total Cost                            2868.75
Average Cost                             0.9201
=== Confusion Matrix ===

   a   b   c   d   e   <-- classified as
 229  20  15   8   1 |   a = 1
 185   7  40  30   2 |   b = 2
 231  22 128 118   4 |   c = 3
 270  50 151 562  57 |   d = 4
 202  41  81 519 145 |   e = 5
```

Chosen cost matrix generates such results as the cost matrix assumes conservative approach to costs, especially extreme false positives as discussed earlier. Total cost is significally lower

while the predictive power of the model settles at the benchmark therefore a potential model for solely cost based approach.

Review text:

```
Correctly Classified Instances        1391               44.6119 %
Incorrectly Classified Instances      1727               55.3881 %
Kappa statistic                          0.1831
Total Cost                            5015
Average Cost                             1.6084
=== Confusion Matrix ===

   a   b   c   d   e   <-- classified as
  64   7  14 131  57 |   a = 1
  22   3  17 175  47 |   b = 2
   3   0  41 358 101 |   c = 3
   3   1  33 702 351 |   d = 4
   0   0  11 396 581 |   e = 5
```

Correctly classified instances is significally higher than benchmark, but lower than the best title model of 47.2739 % indicating that the review title might include all the relevant words for predicting comparing to review text. Confusion matrix implies that possibly some similar neutral words used in good and excellent reviews are used in bad and medium rating review texts, therefore skewing those towards good and excellent ratings.

Attribute selection - >0:

```
Correctly Classified Instances        1440               46.1835 %
Incorrectly Classified Instances      1678               53.8165 %
Kappa statistic                          0.2212
Total Cost                            4440
Average Cost                             1.424
=== Confusion Matrix ===

   a   b   c   d   e   <-- classified as
 117   8  24  89  35 |   a = 1
  41  13  34 138  38 |   b = 2
  10   6  54 347  86 |   c = 3
   9   6  53 679 343 |   d = 4
   4   0  22 385 577 |   e = 5
```

By selecting only attributes with information gain with respect to the class improves the model when working with review text that usually includes more words than review title, as well as more irrelevant words for the signal.

CSC (all attributes):

```
Correctly Classified Instances          914               29.3137 %
Incorrectly Classified Instances       2204               70.6863 %
Kappa statistic                           0.0994
Total Cost                             2628
Average Cost                              0.8428
 === Confusion Matrix ===

   a   b   c   d   e   <-- classified as
 261   0   4   8   0 |   a = 1
 235   0  11  18   0 |   b = 2
 337   1  42 123   0 |   c = 3
 406   8  66 610   0 |   d = 4
 215   5  61 706   1 |   e = 5
```

Correctly classified instances is below the benchmark level, however the total cost is the lowest among the introduced models, indicating a potential business case for the model

2.

Modeling how influental the review will be with interest of 15 or more helpful votes recieved. A new binary attribute is introduced to indicate wether the review have received 15 or more helpful votes (1) or not (0).

Texts are mined by indetifying which words appears with a frequency of 5 or more, and using a machine learning algo to construct a predictive model for the bianary variable value in question.

RandomForest is used for robustness of the model. The model uses cross-validtion of 10 fold to create a robust prediction model.

SMOTE is included in the comparisation of the model to show robustness and to minimize sample size bias, which is apperant in the balance of binary variable values - 3106 (0) 12 (1). A 10000% increase (100x) is applied to binary value (1).

Only attributes with information gain with respect to the class (>0) improves the text model as seen in the section 1 of the report and hence is also used in the section 2 of the report.

Analysis

ZeroR, which works as a benchmark, as it assigns the instance to a binary variable that is most likely to appear in the dataset a.k.a mode (0). ZeroR produces 99.6151 % of correctly classified instances.

Title:

```
Correctly Classified Instances        3106               99.6151 %
Incorrectly Classified Instances        12                0.3849 %
=== Confusion Matrix ===

    a    b   <-- classified as
 3105    1 |    a = 0
   11    1 |    b = 1
```

Model offers no value when modelling with the original dataset.

(with smote):

```
Correctly Classified Instances        4306               99.7221 %
Incorrectly Classified Instances        12                0.2779 %
=== Confusion Matrix ===

    a    b   <-- classified as
 3103    3 |    a = 0
    9 1203 |    b = 1
```

With synthetic additional data, the predictive power increases slightly, justifying the use of SMOTE.

Text:

```
Correctly Classified Instances        3106               99.6151 %
Incorrectly Classified Instances        12                0.3849 %
=== Confusion Matrix ===

    a    b   <-- classified as
 3106    0 |    a = 0
   12    0 |    b = 1
```

Model offers no value when modelling with the original dataset.

(with smote):

```
Correctly Classified Instances        4315               99.9305 %
Incorrectly Classified Instances         3                0.0695 %
=== Confusion Matrix ===

    a    b   <-- classified as
 3106    0 |    a = 0
    3 1209 |    b = 1
```

Synthetic instances improves the model predictive power dramatically and therefore offers high value. Comparing title (99.7221%) and text results (99.9305 %), the difference could be explained due to the reviews text's explanitary nature of the review therefore prompting a helpful vote or not.

3.

Modeling how influental the review author is in real life by assigning the instance to a new binary variable, where a review author has visited more than 15 cities (1) or not (0).

Variables [num_heplful_votes, via_mobile, author_num_helpful_votes, author_num_reviews] are used to construct the predictive model, as these variables could be seen as some of the explanatory variables for author's influence. Variable via_mobile is translated to a bianary variable.

A comparison of J48 and RandomForest as well as Neural Network is used to construct the potential modes for the model. Also the time complexity of the models is considered.

Also, models uses cross-validtion of 10 fold to create a robust prediction model.

Analysis

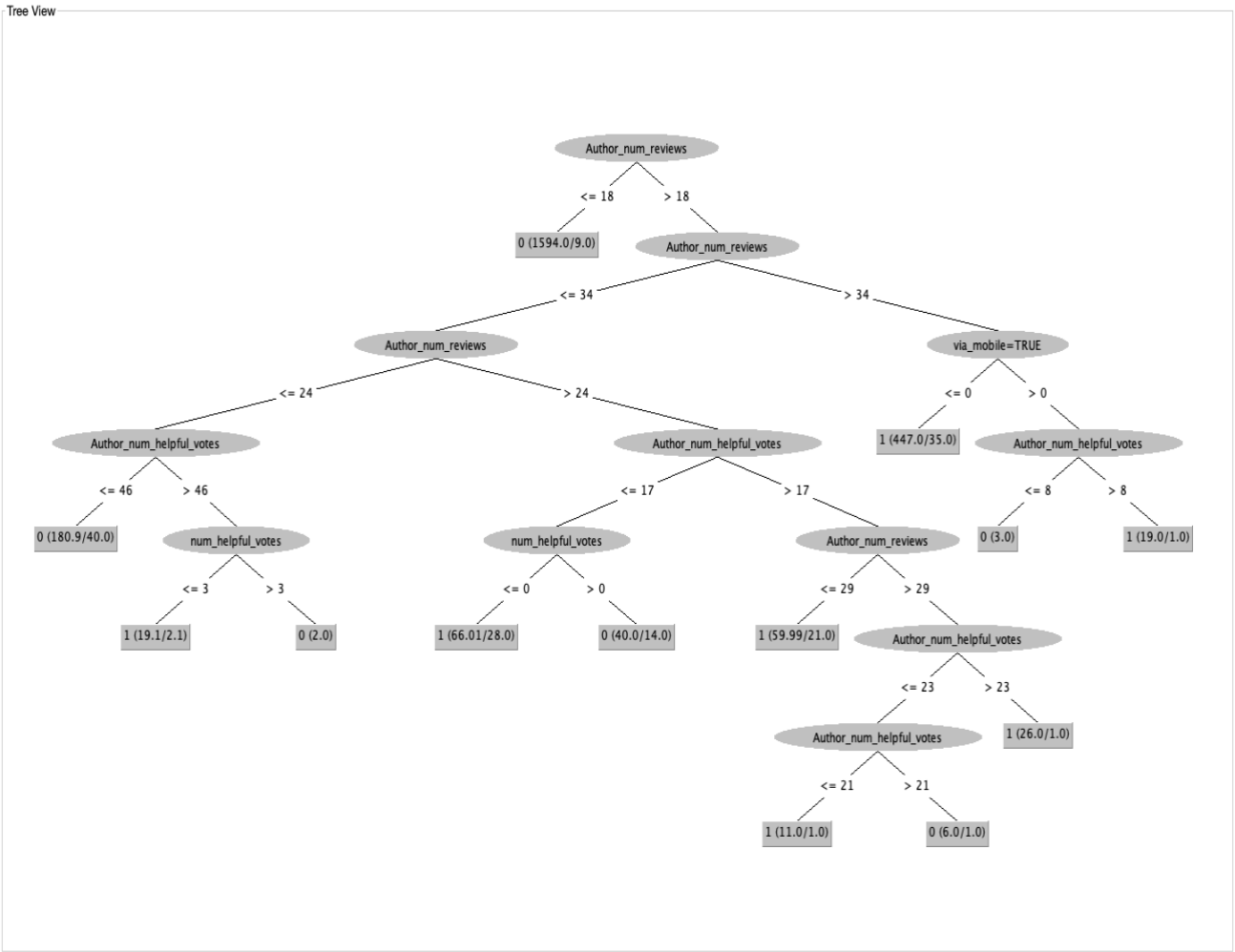
ZeroR, which works as a benchmark, as it assigns the instance to a binary variable that is most likely to appear in the dataset a.k.a mode a (0). ZeroR produces 74.8181 % of correctly classified instances.

```
Correctly Classified Instances        1851               74.8181 %
Incorrectly Classified Instances       623               25.1819 %
=== Confusion Matrix ===

    a    b   <-- classified as
 1851    0 |    a = 0
  623    0 |    b = 1
```

J48 (time taken to build model: 0.12 seconds):

```
Correctly Classified Instances        2287               92.4414 %
Incorrectly Classified Instances       187                7.5586 %
=== Confusion Matrix ===

    a    b   <-- classified as
 1736  115 |    a = 0
   72  551 |    b = 1
```

Model produces significally higher correctly classidfied instances percentage than the benchmark, indicating a use for model in the business case.

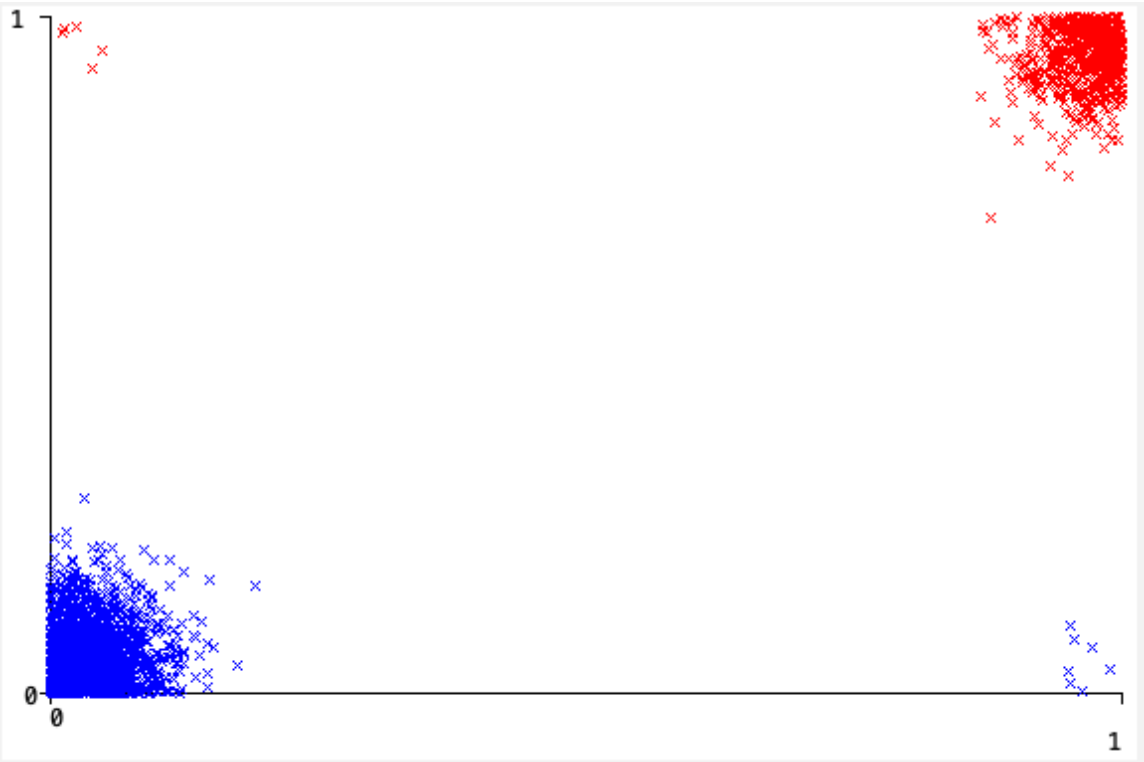


J48 decision tree viualization. The model has divided the attributes to sub nodes according to it's evaluation of the best prediction as such.

RandomForest (time taken to build model: 1.2 seconds):

```
Correctly Classified Instances        2294               92.7243 %
Incorrectly Classified Instances       180                7.2757 %
=== Confusion Matrix ===

    a    b   <-- classified as
 1751  100 |    a = 0
   80  543 |    b = 1
```

RandomForest model improves sligthly the predictive power compared to the J48, but the model takes consideribly more time in comparision to J48.



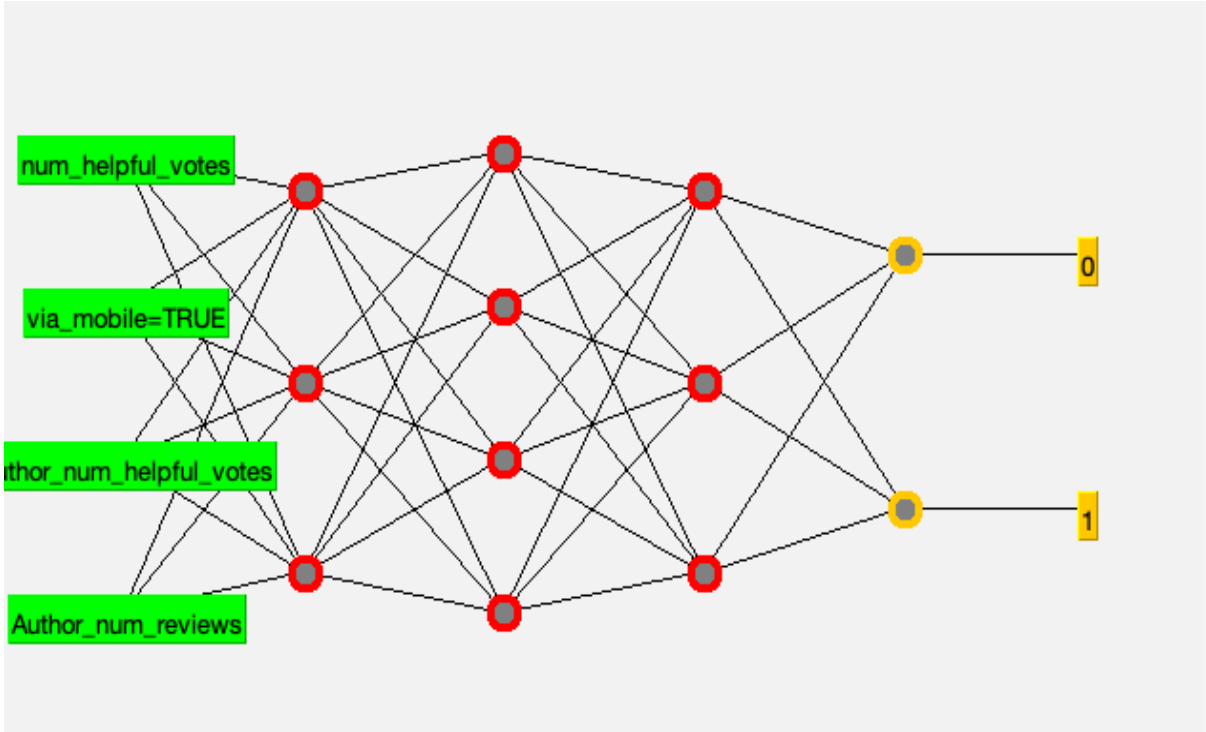
Visualization of the predicted values (y-axis) and the actual values (x-axis). No clear expalanation for the incorrect predictions is found.

Neural Network:

```
Correctly Classified Instances        2280               92.1584 %
Incorrectly Classified Instances       194                7.8416 %
=== Confusion Matrix ===

    a    b   <-- classified as
 1731  120 |    a = 0
   74  549 |    b = 1
```

The predictive power of the neural nerwork model is the worst among comparision.

Model is trained 500 times using 3,4,3 nodes. The number of training and arragnment of nodes could be experimented further to explore possible improvements for the model. Also, some other variables might be justified to include in the analysis and hence requires evaluation.



A visualization of the neural network model.